



Network Programming (IE2102)
2nd Year, 1st Semester

Assignment

**Design and Implementation of a Secure Multiprocessor
TCP Application Server with Custom Protocol and
Session Management**

Submitted to
Sri Lanka Institute of Information Technology

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

<<2026/4/10>>

Important:

Remember

- *Plagiarism is considered as a serious crime against intellectual property in academic environments.*
- *By including the declaration page mentioned below, you abide by the laws of the institute to accept 0 marks for the report, in case you get caught attempting plagiarism.*

Annexure 1

General Formatting

1. Paper size: A4
2. Contents font & size: Times New Roman, size 12
3. Chapter headings/sub headings font & size: Times New Roman Bold, size 12
4. Page numbering:
 - Cover page should not contain any number
 - Declaration page to table of figures page should be numbered using Roman numbers at bottom center of the page
 - Content pages, references page(s) and annexes (if any) should be numbered using Arabic numbers at the bottom center of the page
5. Margins: 1 inch margins on left, right, top and bottom
6. Line spacing: 1 line spacing

Figure Numbering:

1. Each figure should be numbered at bottom center of the figure along with a figure name.
2. Figure number should start with word 'Figure' followed by the 'main chapter number' and 'figure number within the chapter' separated by a 'period sign (.)' and followed by a 'colon (:)'.

For example,

Figure 2.6: XYZ Connector

Declaration Page:

Declaration page should contain the following. Reports that does not contain this declaration 'AS IS' will not be accepted as valid reports and will be awarded 0 marks.

Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

a. Hostname configuration

```
(manith@LAPTOP-U3BF6G0N)-[/mnt/c/Users/Manith Banula/IE2102_Assignment]
$ hostname
LAPTOP-U3BF6G0N
```

- Display the configured hostname of the server machine

b. Network configuration

```
(manith@LAPTOP-U3BF6G0N)-[/mnt/c/Users/Manith Banula/IE2102_Assignment]
$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:34:18:8e:12 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.26.83.240 netmask 255.255.240.0 broadcast 172.26.95.255
    inet6 fe80::215:5dff:fe85:25d2 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:85:25:d2 txqueuelen 1000 (Ethernet)
    RX packets 27389 bytes 39100433 (37.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13719 bytes 1004746 (981.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 3344 (3.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 3344 (3.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- Displays network interfaces and IP address of the server.

c. Server running

```
(manith@LAPTOP-U3BF6G0N)-[/mnt/c/Users/Manith Banula/IE2102_Assignment]
$ ss -tulnp | grep 50776
tcp LISTEN 0 10 0.0.0.0:50776 0.0.0.0:* users:(("server_5776",pid=790,fd=3))
```

- Confirms that the server is ready to receive connections on the assigned port

d. Successful REGISTER and LOGIN

```
> REGISTER ranasinghe 1122
[SERVER] OK 200 SID:1024 User Registered

> LOGIN ranasinghe 1122
[SERVER] OK 200 SID:1024 TK17759889084084
[+] Token saved: TK17759889084084

> LOGOUT ranasinghe 1122
[SERVER] OK 200 SID:1024 Logged Out
[*] Logged out successfully
```

- Demonstrates successful user registration and login.

e. Failed login + logout proof

```

> LOGIN ranasinghe 1234
[SERVER] ERR 200 SID:1024 Auth Fail

> LOGIN ranasinghe 1234
[SERVER] ERR 200 SID:1024 Auth Fail

> LOGIN ranasinghe 1234
[SERVER] ERR 200 SID:1024 Auth Fail

> LOGIN ranasinghe 1234
[SERVER] ERR 200 SID:1024 Account Locked

```

- Highlights defense against brute-force attacks through temporary account locking

f. Multiple forked processes

```

(manith@LAPTOP-U3BF6G0N)~/mnt/c/Users/Manith Banula/IE2102_Assignment
$ ps aux | grep server
root      7  0.0  0.0   3120  1924 ?        Sl   18:04   0:00 plan9 --c
ontrol-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
manith    2419 0.0  0.0   2824  1408 pts/0    S+   18:22   0:00 ./server
manith    2952 0.0  0.0   2824  1792 pts/0    S+   19:11   0:00 ./server
manith    3113 0.0  0.0   6612  2048 pts/3    S+   19:27   0:00 grep --co
lor=auto server

```

- The system creates separate processes via `fork()` to handle multiple clients simultaneously

g. Log file entries

```
(manith@LAPTOP-U3BF6G0N)-[/mnt/c/Users/Manith Banula/IE2102_Assignment]
$ cat server_IT24102776.log
[Fri Apr 10 19:11:17 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:REGISTER D
ETAILS:manith
[Fri Apr 10 19:11:52 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOGIN_OK D
ETAILS:manith
[Fri Apr 10 19:12:03 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOGOUT DET
AILS:OK
[Fri Apr 10 19:14:29 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOGIN_FAIL
DETAILS:manith
[Fri Apr 10 19:14:49 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOGIN_FAIL
DETAILS:manith
[Fri Apr 10 19:15:05 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOGIN_FAIL
DETAILS:manith
[Fri Apr 10 19:15:21 2026] PID:2952 IP:127.0.0.1 PORT:39866 EVENT:LOCKED DET
AILS:manith
[Fri Apr 10 19:47:31 2026] PID:3614 IP:127.0.0.1 PORT:49970 EVENT:REGISTER D
ETAILS:banula
[Sun Apr 12 15:44:01 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:REGISTER DE
TAILS:manith
[Sun Apr 12 15:44:10 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOCKED DETA
ILS:manith
[Sun Apr 12 15:44:56 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:REGISTER DE
TAILS:ranasinghe
[Sun Apr 12 15:45:08 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOGIN_OK DE
TAILS:ranasinghe
[Sun Apr 12 15:45:22 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOGOUT DETA
ILS:OK
[Sun Apr 12 16:08:23 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOGIN_FAIL
DETAILS:ranasinghe
[Sun Apr 12 16:08:43 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOGIN_FAIL
DETAILS:ranasinghe
[Sun Apr 12 16:08:54 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOGIN_FAIL
DETAILS:ranasinghe
[Sun Apr 12 16:09:13 2026] PID:853 IP:127.0.0.1 PORT:54062 EVENT:LOCKED DETA
ILS:ranasinghe
```

- Server logs capture key information including time, client IP address, PID, user, command executed, and result

h. Proper child cleanup (no zombie processes)

```
(manith@LAPTOP-U3BF6G0N)-[/mnt/c/Users/Manith Banula/IE2102_Assignment]
$ ps aux | grep Z
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
manith    1297  0.0  0.0   6612  2048 pts/3    S+   16:20   0:00 grep --co
lor=auto Z
```

- No zombie processes exist, confirming proper SIGCHLD handling.

